Remote Sensing Image Scene Classification Using Multiscale Feature Fusion Covariance Network With Octave Convolution

Lin Bai, *Member, IEEE*, Qingxin Liu, Cuiling Li, Zhen Ye, Meng Hui, *Member, IEEE*, and Xiuping Jia, *Fellow, IEEE*

***Abstract*— In remote sensing scene classification (RSSC), features can be extracted with different spatial frequencies where high-frequency features usually represent detailed information and low-frequency features usually represent global structures. However, it is challenging to extract meaningful semantic information for RSSC tasks by just utilizing high- or low-frequency features. The spatial composition of remote sensing images (RSIs) is more complex than that of natural images, and the scales of objects vary significantly. In this article, a multiscale feature fusion covariance network (MF^{2}CNet) with octave convolution (Oct Conv) is proposed, which can extract multifrequency and multiscale features from RSIs. First, the multifrequency feature extraction (MFE) module is used to obtain fine-grained frequency features by Oct Conv. Then, the features of different layers in MF^{2}CNet are fused by the multiscale feature fusion (MF^2) module. Finally, instead of using global average pooling (GAP), global covariance pooling (GCP) extracts high-order information from RSIs to capture richer statistics of deep features. In the proposed MF^{2}CNet, the obtained multifrequency and multiscale features can effectively improve the performance of CNNs. Experimental results on four public RSI datasets show that MF^{2}CNet has advantages in RSSC over current state-of-the-art methods. The source codes of this method can be found at https://github.com/liuqingxin-chd/MF2CNet.**

***Index Terms*— Convolutional neural networks (CNNs), covariance pooling, multiscale feature fusion (MF^2), octave convolution (Oct Conv), scene classification.**

I. Introduction

REMOTE sensing is a critical technique for Earth observation from satellites or spacecraft. Currently, the application of remote sensing involves many fields, such as urban planning and environmental monitoring [1]. The rapid development of remote sensing, electronic information, and imaging technologies provides us with a large number of remote sensing images (RSIs), such as super large FMoW dataset [2], which are highly useful for deep learning development and validation [3]. Compared to pixel- or object-level classification, remote sensing scene classification (RSSC) is more complex and difficult. For example, there are many different types of ground objects with significant size variations in RSIs, while many of them are not in the center of the images. The RSIs in Fig. 1 are selected from the NWPU-RESISC45 dataset [4]. It can be seen that the size of ground objects varies greatly in different scenes. These issues have made feature representation learning in RSSC tasks extremely difficult. As a result, it is critical to develop an efficient scene categorization technique for RSIs [5].

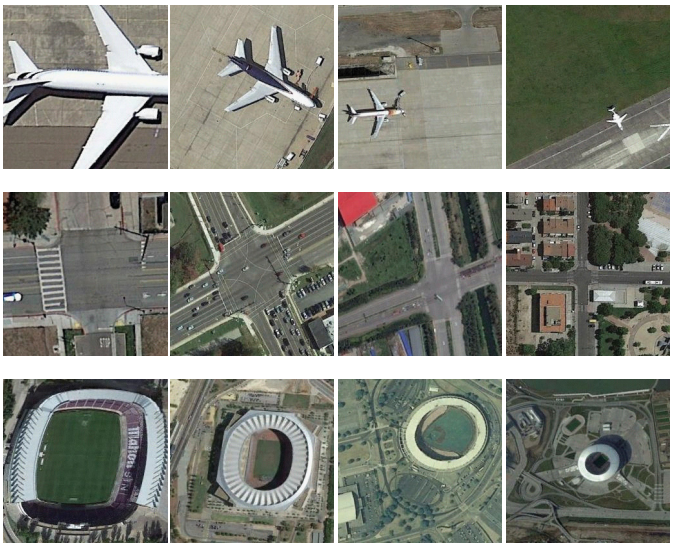

Fig. 1. Multiscale objects of RSI dataset in the following scenes: airplane (the first row), intersection (the second row), and stadium (the third row).

A variety of approaches have evolved for RSSC tasks during the last few decades [6]. Early efforts are mainly concentrated on handcrafted feature-based methods for RSIs. Most of these methods are designed according to engineering skills, such as scale-invariant feature transform (SIFT) [7], local binary pattern (LBP) [8], the Gabor feature [9], and histogram of

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oriented gradients (HOG) [10]. To represent the whole scene of RSIs, some other methods [11] by constructing global image features have been proposed, such as the well-known bag-of-visual-words (BoVW) [12] and spatial pyramid matching (SPM) [13]. However, handcrafted features may be limited in high-level semantic information extraction [4]. Therefore, deep feature-based methods have recently been studied for RSSC.

With the development of deep learning in the field of computer vision, many deep learning methods were introduced in remote sensing [14]. Convolutional neural networks (CNNs), such as ResNet [15] and VGG16 [16], were utilized in RSSC tasks to improve classification performance. Transfer learning [17], [18] were used to prevent the networks from overfitting. Some methods based on knowledge distillation [19] were used to learn the features from RSIs in a lightweight way. Metric learning [20] was also used to improve the classification accuracies by improving the loss function. What is more, EfficientNet [21] was utilized to extract semantic information in an efficient way. GhostNet [22] as a lightweight network was utilized to classify RSIs. The vision transformer [23] used an attention mechanism to extract features from RSIs. In addition, deep CNN is also used to jointly process multisource remote sensing data [24], [25]. Although the above methods based on deep learning show some effectiveness, their classification performances are sensitive to different sizes of ground objects and complex scenes.

For extracting multiscale features and reducing spatial redundancy, Chen *et al.* [26] proposed octave convolution (Oct Conv) to improve the performance of CNN. Because Oct Conv can capture multifrequency feature representation, Oct Conv is used to learn multifrequency feature representation in the field of hyperspectral image classification [27], [28]. Only using the high-level semantic information extracted by the last convolution layer may lose some smaller objects, which is not conducive to the classification task for complex scenes. On the other hand, some researchers use high-order (second-order) pooling methods [29] to retain deep semantic information from RSIs. He *et al.* [30] proposed a skip-connected covariance network to learn second-order information (variance) for RSSC. Wang *et al.* [31] utilized invariant deep compressible covariance pooling (IDCCP) to extract the complex features in RSIs. These covariance pooling methods depend heavily on eigendecomposition (EIG) or singular value decomposition (SVD). However, implementation support for EIG or SVD on the graphics processing unit (GPU) is limited and even slower than on the central processing unit (CPU) [32], [33].

It is challenging to extract comprehensive multiscale features from RSIs due to their variable objects and complex spatial structure. In this article, we propose a multiscale feature fusion covariance network (MF^2CNet) to extract multifrequency and multiscale features for RSSC tasks. The multifrequency feature extraction (MFE) module is designed to extract multifrequency features with Oct Conv by strengthening the information interaction between high frequency and low frequency. Specifically, inspired by the advantages of layerwise feature fusion methods [30], [34], the feature maps of different positions in the network are extracted for feature fusion. To fuse information from different layers containing varying spatial frequency features, a multiscale feature fusion (MF^2) module is proposed. The MF^2 module not only fuses the multiscale features from different layers but also fuses the high- and low-frequency information of each layer. Furthermore, in order to capture richer statistics of multifrequency and multiscale features, the global average pooling (GAP) is replaced by GPU-friendly global covariance pooling (GCP) module to downsample the feature map. Compared with GAP that only extracts first-order information (mean), the GCP module can extract second-order information (covariance). Moreover, the GCP module does not use the matrix inverse operation, so it can be implemented on GPU. By comparing a large number of recently proposed RSSC methods, we further analyze and discuss the proposed MF^2CNet by a series of ablation experiments. Besides that, we also use the visualization method to analyze the classification performance of the proposed network.

The significant contributions can be summarized as follows.

1) In this article, an MFE module based on Oct Conv is proposed in perspective of the effect of spatial frequency features on the expression ability of CNNs. MFE, as distinct to interlayer feature extraction, adopts a convolution layer to implement fine-grained MFE.
2) An MF^2 module is proposed to fuse information from different layers with various spatial-frequency features. The MF^2 module can supplement some valuable information extracted from the shallow level to the deep semantic information to enhance the feature expression ability for complex scenes.
3) Using the GCP module with rich statistical information instead of the conventional GAP in the network can effectively preserve the fused semantic information. Furthermore, the GCP module in our network is more GPU-friendly.
4) MF^2CNet is an end-to-end network, which can be well used for RSSC tasks. Compared with a large number of state-of-the-art methods, the multifrequency and multiscale features extracted by MF^2CNet can significantly improve the accuracies of RSSC.

The remainder of this article is organized as follows. Section II introduces the related work in the field of RSSC and the innovation of our network. Section III recommends the proposed MF^2CNet and its essential parts in detail. The experiments and analyses on four RSI datasets of the proposed network are shown in Section IV. Section V discusses the influences of various factors in MF^2CNet. Finally, we summarize this article in Section VI and plan for future studies in general.

II. Related Work

In the past decades, researchers have done many works to handle the RSSC tasks. However, due to the variable objects and complex spatial structure of RSIs, it is a challenge to design RSSC methods with high performance. According to the features extracted by the existing RSSC methods, we classify them into two categories: 1) deep feature-based methods and 2) multiscale feature-based methods.

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A. Deep Feature-Based Methods

Unlike handcrafted feature-based methods, deep feature-based methods can automatically learn features and extract rich information from original data. Since the methods based on deep features have strong feature learning abilities, the classification accuracies can be improved in RSSC tasks. Sun *et al.* [35] proposed a gated bidirectional network to fuse multilayer features. Fang *et al.* [36] combined frequency-domain with the traditional CNN architecture for RSSC. Liu *et al.* [37] utilized the hierarchical Wasserstein distance to construct a new loss function that improves the performance of the network. Yuan *et al.* [38] proposed a method based on rearranged local features in RSIs. Penatti *et al.* [39] evaluated the generalization power of deep features. Minetto *et al.* [40] created an ensemble of two CNNs for geospatial land classification. Gong *et al.* [41] adopted two simple and effective strategies to extract CNN features. Anwer *et al.* [42] investigated binary patterns encoded CNN and different deep network fusion architectures. Wang *et al.* [43] proposed an end-to-end attention recurrent convolutional network (ARCNet). Tang *et al.* [44] designed an attention consistent network (ACNet) to extract the features from RSIs. In summary, the different architectures of the network were designed to capture the features from RSIs, and the classification results have made some improvements. However, the performance of the above methods needs to be further improved when dealing with some variable objects, complex scenes, and different spatial resolutions.

B. Multiscale Feature-Based Methods

To deal with the above problems, researchers have proposed a large number of methods to extract multiscale features from RSIs. First, multiscale features can be learned by scaling the resolutions of the input RSIs. Xie *et al.* [45] proposed a scale-free CNN (SF-CNN) to preserve the critical information in RSIs during the image resizing process. Wang *et al.* [46] utilized a multiscale representation for scene classification, which is conducted by designing a global–local two-stream architecture. Besides that, some more fine-grained methods were proposed to learn multiscale features in RSIs. Zhang *et al.* [47] used dilated convolution with different dilation and mutilation pooling modules to learn multiscale features. Bai *et al.* [48] proposed a multiscale depthwise convolution and efficient spatial pyramid channel attention for RSSC. In addition, feature fusion methods of different scales from different layers are proposed. He *et al.* [49] used multilayer stacked covariance pooling (MSCP) to construct multilayer feature maps of RSIs. Lu *et al.* [34] proposed a feature aggregation CNN (FACNN) to extract multiscale scene representation. Cao *et al.* [50] utilized self-attention-based deep feature fusion (SAFF) to aggregate multiscale features of RSIs. Moreover, some methods not only scale the resolutions of the input image but also fuse the multiscale features from different layers. Li *et al.* [51] proposed a multiscale improved Fisher kernel (MIFK) coding method and a feature fusion framework to extract multiscale features for RSSC.

III. Proposed Method

Inspired by the methods of fusing multiscale features from different layers, we propose a novel MF^2 module based on high- and low-frequency information extracted by Oct Conv. The MF^2 module fuses feature maps from different layers and fuses different spatial-frequency information on each layer. The multiscale features from different layers of the network are extracted to obtain multiscale fusion information, and then, the feature maps are concatenated. Furthermore, GCP is used to preserve multifrequency and multiscale information. Compared with GAP, GCP introduces richer statistic information in the downsampling step. In addition, the GCP does not use the matrix inverse operation, so it can be implemented on GPU. The MFE module with Oct Conv divides the feature maps into two branches: high- and low-frequency features. In addition, information interaction is achieved by using high- and low-frequency information. Oct Conv extracts multifrequency features to zoom out the sizes of low-frequency components, which can save computational cost and resource occupation.

A. Overall Architecture

The overall architecture of the proposed MF^{2}CNet is shown in Fig. 2, which mainly contains three parts, i.e., the MFE module, the MF^2 module, and the GCP module. First, the input RSIs with the resolution of $224 \times 224 \times 3$ are processed on the first convolution layer and the maximum pooling layer. Then, the feature maps are sent to the MFE module. For the multifrequency feature extracting, the Oct Conv is utilized to extract multifrequency features in the MFE module. The MFE module has four stages, each of which is composed of multiple Oct-bottlenecks. The detailed structure of the bottleneck block is shown in Fig. 2(a). Oct-bottleneck is divided into three cases: Oct-bottleneck (F), Oct-bottleneck (M), and Oct-bottleneck (L), which represent the first, middle, and last Oct-bottlenecks in the MFE module, respectively. For example, the high-frequency feature HF_{Mi} and the low-frequency feature LF_{Mi} are inputted to Oct-bottleneck (M); and the high-frequency feature HF_{Mo} and low-frequency feature LF_{Mo} are outputted. In Oct-bottlenecks, the ResNet identity mappings are used to construct residual structures to prevent network degradation. In addition, batch normalization [52] and ReLU activations [53] are also used in Oct-bottleneck. Oct Conv 1×1 is used to expand or reduce the channel number of the feature map. Then, feature maps of different scales are sent to the MF^2 module for fusion to obtain the multiscale features. Specifically, feature maps from stages 2 to 4 are concatenated in the MF^2 module after downsampling and dimensionality reduction. Then, the GCP module replaces GAP to obtain more rich information. Finally, the semantic information generated on the fully connected layer serves classifiers.

B. Multifrequency Feature Extraction Module

In RSIs, low-frequency features usually contain global information, and high-frequency features usually contain details. Similarly, the feature maps of the convolutional layer's output can be regarded as a mixture of high- and low-frequency

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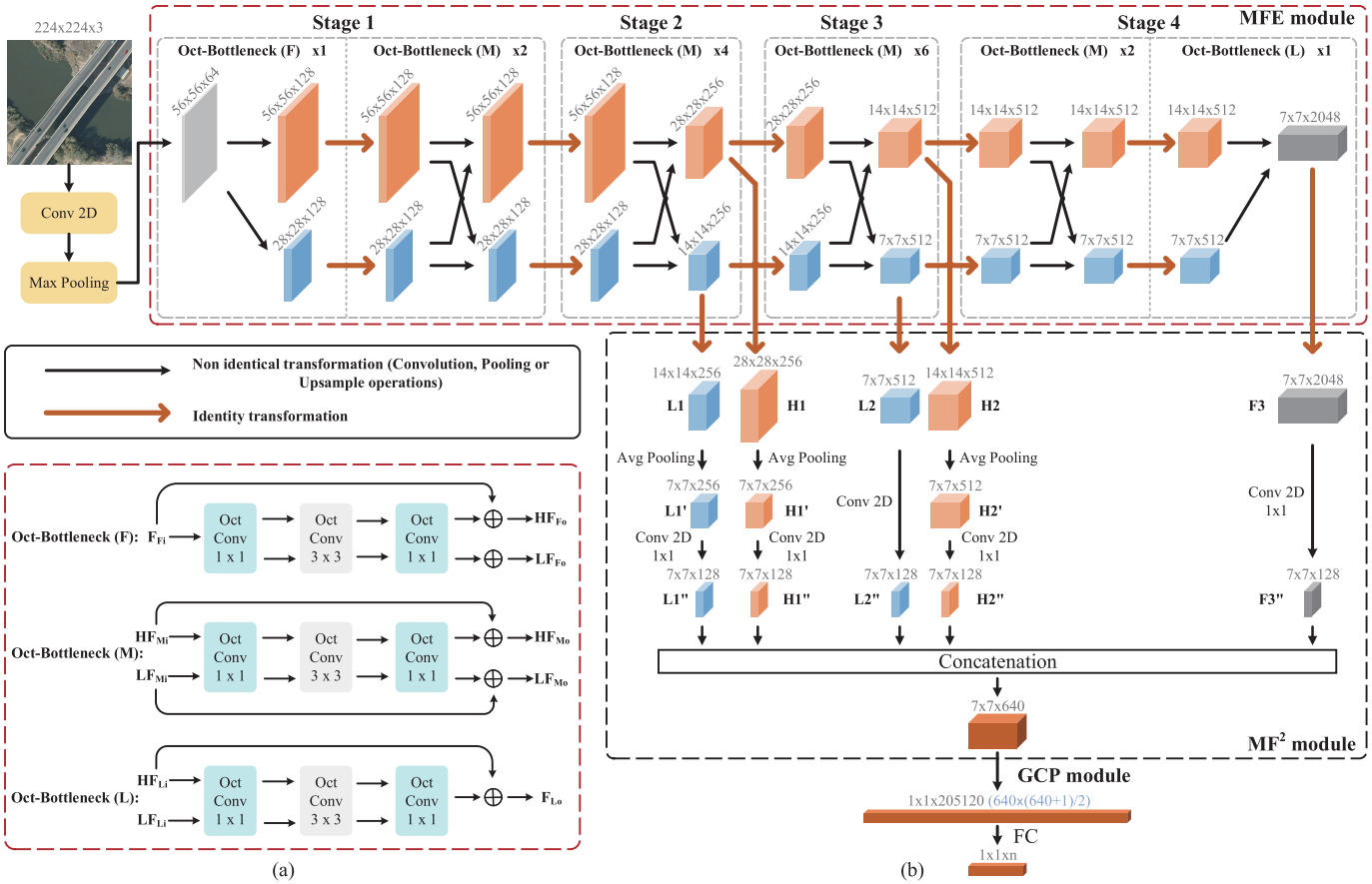

Fig. 2. General structure of MF^{2}CNet and the internal structure of Oct-Bottleneck. (a) Internal structure of Oct-Bottleneck, which is divided into three cases: F (first), M (middle), and L (last). (b) General structure of MF^{2}CNet includes the MFE module, the MF^2 module, and the GCP module. The number of scene categories in RSSC is represented by n.

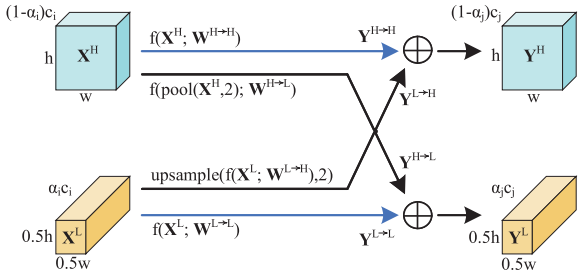

Fig. 3. Structure of Oct Conv [26].

TABLE I
NUMBER OF PARAMETERS AND FLOPS OF STANDARD CONV AND PARTS OF OCT CONV AND OCT CONV

	Parameters	FLOPs
Standard Conv	$k^2c_ic_j$	$hwk^2c_ic_j$
H → H	$k^2(1-\alpha_i)c_i(1-\alpha_j)c_j$	$hwk^2(1-\alpha_i)c_i(1-\alpha_j)c_j$
H → L	$k^2(1-\alpha_i)c_i\alpha_jc_j$	$\frac{1}{4}hwk^2(1-\alpha_i)c_i\alpha_jc_j$
L → H	$k^2\alpha_ic_i(1-\alpha_j)c_j$	$\frac{1}{4}hwk^2\alpha_ic_i(1-\alpha_j)c_j$
L → L	$k^2\alpha_ic_i\alpha_jc_j$	$\frac{1}{4}hwk^2\alpha_ic_i\alpha_jc_j$
Oct Conv	$k^2c_ic_j$	$hwk^2c_ic_j[1-\frac{3}{4}(\alpha_i+\alpha_j-\alpha_i\alpha_j)]$

information. In order to extract multifrequency features efficiently and reduce spatial redundancy, Chen *et al.* [26] proposed Oct Conv to improve the performance of convolution. Oct Conv decomposes the feature maps into two parts: high- and low-frequency components. The spatial resolutions of low-frequency feature maps are half of the high-frequency ones, so the costs of storage and calculation will be reduced effectively. Besides that, pooling and upsampling are utilized to expand the receptive fields of convolution without parameters. Therefore, we designed an MFE module with Oct Conv to extract multifrequency features from RSIs.

The structure of Oct Conv is shown in Fig. 3. The input of Oct Conv has two parts: the high-frequency feature map $\mathbf{X}^H$ and the low-frequency feature map $\mathbf{X}^L$. The height and the width of $\mathbf{X}^L$ are half of ones of $\mathbf{X}^H$. The channel numbers of $\mathbf{X}^H$ and $\mathbf{X}^L$ are both controlled by α_i, where $\alpha_i \in [0, 1]$. Then, $\mathbf{Y}^{H\to H}$ and $\mathbf{Y}^{H\to L}$ are obtained by $f(\mathbf{X}^H; \mathbf{W}^{H\to H})$ and $f(\text{pool}(\mathbf{X}^H, 2); \mathbf{W}^{H\to L})$, respectively. $\mathbf{Y}^{L\to H}$ and $\mathbf{Y}^{L\to L}$ are obtained by upsample$(f(\mathbf{X}^L; \mathbf{W}^{L\to H}), 2)$ and $f(\mathbf{X}^L; \mathbf{W}^{L\to L})$, respectively. The next step is the multifrequency information interaction through high- and low-frequency feature fusion. The output of high-frequency feature map $\mathbf{Y}^H$ can be obtained by $\mathbf{Y}^H = \mathbf{Y}^{H\to H} + \mathbf{Y}^{L\to H}$, and the output of low-frequency feature map $\mathbf{Y}^L$ can be obtained by $\mathbf{Y}^L = \mathbf{Y}^{H\to L} + \mathbf{Y}^{L\to L}$. Similarly, the height and the width of $\mathbf{Y}^L$ are half of that of $\mathbf{Y}^H$. The channel numbers of $\mathbf{Y}^H$ and $\mathbf{Y}^L$ are controlled by α_j, where $\alpha_j \in [0, 1]$.

As shown in Table I, we compare the number of parameters and floating-point operations (FLOPs) of standard

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convolution (Standard Conv) and parts of Oct Conv (H → H, H → L, L → H, and L → L) and Oct Conv. The numbers of parameters of Standard Conv and Oct Conv are the same. Moreover, the FLOPs of Oct Conv are smaller than the one of Standard Conv. When α_i and α_j are equal to 0.5, the FLOPs of Oct Conv is about 44% of that of Standard Conv.

C. Multiscale Feature Fusion Module

In CNN, different convolution layers have different receptive fields and contain different spatial-scale information. In order to better distinguish the categories of different objects, the multiscale features are extracted to improve the classification performance of the network. However, the general CNN-based feature fusion approaches only consider fusing interlayer features, while the fine-grained spatial frequency information also needs to be considered for feature fusion after using Oct Conv. Therefore, we proposed the MF² module to fuse the high- and low-frequency features from different layers.

As the scales of the feature maps from stage 1 are too large, too much information will be lost if pooling is operated by 7 × 7. To solve this problem, the high- and low-frequency features are extracted from stage 2 to stage 4 for MF². The sizes of **H1**, **L1**, **H2**, **L2** and **F3** are shown in Fig. 2. For example, the size of **F3** is 7 × 7 × 2048. Then, the MF² is accomplished by concatenation, which requires feature maps with the same scale. Finally, the average pooling is implemented to get **H1**′, **L1**′ and **H2**′ from **H1**, **L1** and **H2**.

In order to reduce the computational complexity of GCP and the number of parameters of the whole network, the number of feature map channels needs to be reduced before concatenation. Specifically, the channel numbers of **H1**′, **L1**′, **H2**′, **L2** and **F3** are all reduced to 128 through 1 × 1 convolution. Finally, we concatenated **H1**″, **L1**″, **H2**″, **L2**″ and **F3**″ and fed multiscale features to the GCP module.

D. Global Covariance Pooling Module

For extracting high-order information from feature maps, researchers have done a lot of work. Lin *et al.* [54] proposed a bilinear pooling network for fine-grained classification. Li *et al.* [55] proposed a matrix power normalized covariance (MPN-COV) method to extract second-order information. Furthermore, some methods [56], [57] combine first-order information and second-order information to get more useful information. The acquisition of second-order information consumes more computing costs, so it is necessary to use GPU to accelerate computing. It is well known that the unique square root of any symmetric positive (semi) can be obtained by EIG or SVD. Unfortunately, EIG and SVD cannot be adapted to the GPU. Thus, Li *et al.* [58] proposed an iterative matrix square root normalization of covariance pooling (iSQRT-COV) for parallel computing on GPU to extract second-order information. Existing deep CNNs mostly use GAP to synthesize the last convolution features as the input of the classifier, which captures only first-order statistics and restricts the representation and generalization capabilities of MF²CNet.

The architecture of GCP is shown in Fig. 4. The output feature maps with the size of $h \times w \times d$ from the MF² module

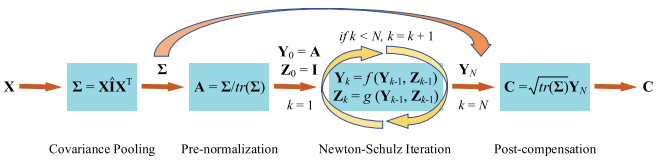

Fig. 4. Architecture of GCP [58].

are transformed with the size of $n \times d$, where $n = h \times w$. First, the covariance matrix Σ is calculated by $\Sigma = \mathbf{X}\hat{\mathbf{I}}\mathbf{X}$, where $\hat{\mathbf{I}} = (1/n)(\mathbf{I} - (1/n)\mathbf{1})$ and $\mathbf{I}$ is the $n \times n$ identity matrix, and $\mathbf{1}$ is the matrix of all ones. Then, the covariance matrix Σ is divided by its trace as $\mathbf{A} = (1/(\mathrm{tr}(\Sigma)))\Sigma$ in the prenormalization step, where tr(·) is the trace of a matrix. This is done so that the subsequent Newton–Schulz iteration can converge [58]. The iterative formulas are given as follows:

$$\mathbf{Y}_k = \frac{1}{2}\mathbf{Y}_{k-1}(3\mathbf{I} - \mathbf{Z}_{k-1}\mathbf{Y}_{k-1}) \tag{1}$$

$$\mathbf{Z}_k = \frac{1}{2}(3\mathbf{I} - \mathbf{Z}_{k-1}\mathbf{Y}_{k-1})\mathbf{Z}_{k-1}. \tag{2}$$

The number of iterations has been determined to be five by cross-validation in [58]. In the postcompensation, $\mathbf{Y}_N$ multiplies the square root of the trace of the covariance matrix to eliminate the adverse effects of the prenormalization by $\mathbf{C} = (\mathrm{tr}(\Sigma))^{1/2}\mathbf{Y}_N$. Finally, the upper triangular matrix of the symmetric matrix obtained from postcompensation is concatenated into a $d(d+1)/2$-dimensional vector, which is sent to the FC layer.

In the backward propagation, the partial derivative of the loss function l relative to the covariance error input matrix is obtained by the gradient associated with network structure in the matrix backpropagation algorithm [59], and the chain rule of the general matrix function is established by first-order Taylor approximation. $(\partial l/\partial \mathbf{C})$ can be calculated, and then, the corresponding gradients $(\partial l/\partial \mathbf{Y}_N)$ and $((\partial l/\partial \Sigma))_{\text{post}}$ are obtained by

$$\frac{\partial l}{\partial \mathbf{Y}_N} = \sqrt{\mathrm{tr}(\Sigma)}\frac{\partial l}{\partial \mathbf{C}} \tag{3}$$

$$\left(\frac{\partial l}{\partial \Sigma}\right)_{\text{post}} = \frac{1}{2\sqrt{\mathrm{tr}(\Sigma)}}\mathrm{tr}\left(\left(\frac{\partial l}{\partial \mathbf{C}}\right)^T \mathbf{Y}_N\right)\mathbf{I}. \tag{4}$$

According to the chain rule of matrix backpropagation and the Newton–Schulz iteration, after a series of operations, $k = N, \ldots, 2$, it can be derived

$$\frac{\partial l}{\partial \mathbf{Y}_{k-1}} = \frac{1}{2}\left(\frac{\partial l}{\partial \mathbf{Y}_k}(3\mathbf{I} - \mathbf{Y}_{k-1}\mathbf{Z}_{k-1}) - \mathbf{Z}_{k-1}\frac{\partial l}{\partial \mathbf{Z}_k}\mathbf{Z}_{k-1} - \mathbf{Z}_{k-1}\mathbf{Y}_{k-1}\frac{\partial l}{\partial \mathbf{Y}_k}\right) \tag{5}$$

$$\frac{\partial l}{\partial \mathbf{Z}_{k-1}} = \frac{1}{2}\left((3\mathbf{I} - \mathbf{Y}_{k-1}\mathbf{Z}_{k-1})\frac{\partial l}{\partial \mathbf{Z}_k} - \mathbf{Y}_{k-1}\frac{\partial l}{\partial \mathbf{Y}_k}\mathbf{Y}_{k-1} - \frac{\partial l}{\partial \mathbf{Z}_k}\mathbf{Z}_{k-1}\mathbf{Y}_{k-1}\right). \tag{6}$$

In prenormalization, $(\partial l/\partial \mathbf{A})$ can be obtained by

$$\frac{\partial l}{\partial \mathbf{A}} = \frac{1}{2}\left(\frac{\partial l}{\partial \mathbf{Y}_1}(3\mathbf{I} - \mathbf{A}) - \frac{\partial l}{\partial \mathbf{Z}_1} - \mathbf{A}\frac{\partial l}{\partial \mathbf{Y}_1}\right). \tag{7}$$

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Fig. 5. Some examples from the UCM21 dataset. 1: agricultural; 2: airplane; 3: baseball diamond; 4: beach; 5: buildings; 6: chaparral; 7: dense residential; 8: forest; 9: freeway; 10: golf course; 11: harbor; 12: intersection; 13: medium residential; 14: mobile home park; 15: overpass; 16: parking lot; 17: river; 18: runway; 19: sparse residential; 20: storage tanks; and 21: tennis courts.

Here, we need to combine the gradient of the loss function l with respect to Σ and backpropagated from the postcompensation layer, which is shown in (4). It can be derived

$$\frac{\partial l}{\partial \Sigma} = -\frac{1}{(\mathrm{tr}(\Sigma))^2}\mathrm{tr}\left(\left(\frac{\partial l}{\partial \mathbf{A}}\right)^T \Sigma\right)\mathbf{I} + \frac{1}{\mathrm{tr}(\Sigma)}\frac{\partial l}{\partial \mathbf{A}} + \left(\frac{\partial l}{\partial \Sigma}\right)_{\text{post}}. \quad (8)$$

Finally, the gradient of the loss function l relative to input matrix $\mathbf{X}$ can be derived as follows:

$$\frac{\partial l}{\partial \mathbf{X}} = \hat{\mathbf{I}}\mathbf{X}\left(\frac{\partial l}{\partial \Sigma} + \left(\frac{\partial l}{\partial \Sigma}\right)^T\right). \quad (9)$$

The parameters in the GCP module can be updated through the backward propagation formulas. For a more detailed derivation, see [58]. Compared with GAP, GCP can better retain semantic information. Most importantly, the GCP module is more suitable for the GPU parallel operation.

IV. Experiments and Analysis

A. Datasets

To verify the classification performance of our proposed method, we choose four widely used datasets.

1) The UC-Merced land-use dataset [60] consists of 21 types of scene classes, and it is abbreviated as UCM21 in this article. Each scene class consists of 100 images with 256 × 256 pixels. The pixel resolution of each image is 1 ft. Fig. 5 shows some examples from the UCM21 dataset.
2) The RSSCN7 dataset [61] consists of seven types of scene classes. Each scene class consists of 400 images with 400 × 400 pixels. Because this dataset is collected in different seasons and weather conditions, the scene images have great diversity and different scales, which brings challenges for RSSC. Fig. 6 shows some examples from the RSSCN7 dataset.
3) The Aerial Image dataset [62] consists of 30 types of scene classes, and it is abbreviated as AID30 in this article. AID30 has 10 000 images in RGB color space, and each scene class of this dataset includes hundreds of images with 600 × 600 pixels. The pixel resolution of each image is from 8 to 0.5 m. Fig. 7 shows some examples from the AID30 dataset.
4) The NWPU-RESISC45 dataset [4] consists of 45 types of scene classes, and it is abbreviated as NWPU45 in this article. Each scene class consists of 700 images with 256 × 256 pixels. The pixel resolution of the image is from 30 to 0.2 m. NWPU45 is a large dataset, which is meant to evaluate networks. Fig. 8 shows some examples from the NWPU45 dataset.

Fig. 6. Some examples from the RSSCN7 dataset. 1: field; 2: forest; 3: grass; 4: industry; 5: parking; 6: resident; and 7: river lake.

Fig. 7. Some examples from the AID30 dataset. 1: airport; 2: bare land; 3: baseball field; 4: beach; 5: bridge; 6: center; 7: church; 8: commercial; 9: dense residential; 10: desert; 11: farmland; 12: forest; 13: industrial; 14: meadow; 15: medium residential; 16: mountain; 17: park; 18: parking; 19: playground; 20: pond; 21: port; 22: railway station; 23: resort; 24: river; 25: school; 26: sparse residential; 27: square; 28: stadium; 29: storage tank; and 30: viaduct.

B. Experimental Settings

We implement all experiments on a personal computer with an Intel CPU i9 9900k, two NVIDIA GeForce RTX 2080Ti GPUs, and 64-GB DDR5 memory. We use the Pytorch platform [63] for experiments. To train the proposed MF^{2}CNet, some parameters of the MFE module are initialized with the pretrained parameters from ImageNet, and the rest parameters need to be obtained through training. The Adam algorithm [64]

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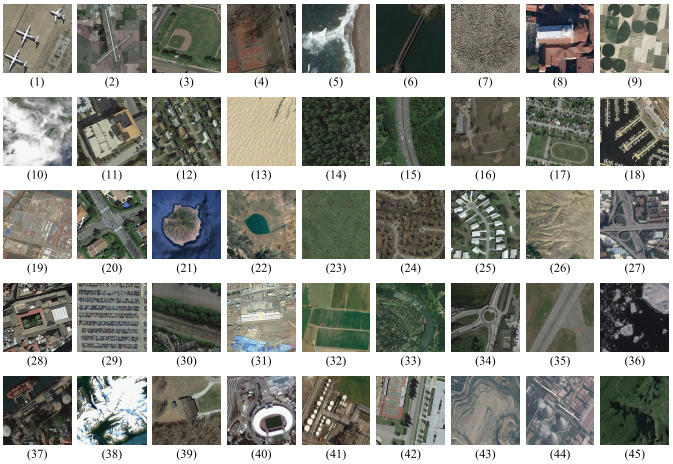

Fig. 8. Some examples from the NWPU45 dataset. 1: airplane; 2: airport; 3: baseball diamond; 4: basketball court; 5: beach; 6: bridge; 7: chaparral; 8: church; 9: circular farmland; 10: cloud; 11: commercial area; 12: dense residential; 13: desert; 14: forest; 15: freeway; 16: golf course; 17: ground track field; 18: harbor; 19: industrial area; 20: intersection; 21: island; 22: lake; 23: meadow; 24: medium residential; 25: mobile home park; 26: mountain; 27: overpass; 28: palace; 29: parking lot; 30: railway; 31: railway station; 32: rectangular farmland; 33: river; 34: roundabout; 35: runway; 36: sea ice; 37: ship; 38: snowberg; 39: sparse residential; 40: stadium; 41: storage tank; 42: tennis court; 43: terrace; 44: thermal power station; and 45: wetland.

is used for optimization with the learning rate of 0.001 at the beginning and then multiplied by 0.1 when the epoch reaches 30, 60, or 90. The weight decay of optimization is set to 0.0005. In addition, the epoch and batch size are set to 120 and 64. The RSIs are processed by data augmentation, such as random rotation, horizontal flip, and vertical flip. Finally, the sizes of the input RSIs are adjusted to 224×224 pixels through the random crop operation.

The overall accuracy (OA) and the confusion matrix (CM) are utilized to value the classification performance of the proposed network. To avoid any bias induced by random sampling, the classification results of the experiments are averaged over ten trials with standard deviation reported.

C. Comparison With State-of-the-Art Methods

Experiment 1 (UCM21 Dataset): First, we do experiments on the UCM21 dataset, which is widely used in RSSC. Same as [60], the training rates (Trs) of the UCM21 dataset are set to 50% and 80%, respectively. We compared the current popular deep learning networks for RSSC. Thirteen approaches are chosen to compare with our method. As shown in Table II, the first three classic CNN [4] are directly implemented for classification. TEX-Net-LF utilizes LBPs encoded CNN models for RSSC. D-CNN uses a metric learning method to classify RSIs, which modifies the loss function in RSSC tasks. FACNN, SAFF, and SCCov adopt layerwise methods by fusing multiscale features to improve classification performance. ARCNet and ACNet employ attention mechanisms. MG-CAP and SCCov improve the pooling layer to extract the distinguishable features. SF-CNN uses original RSIs as input for the RSSC network. EfficientNet-B3-aux [21] uses an auxiliary classification loss function for RSSC. In addition,

TABLE II
COMPARISON OF OAS (%) ON THE UCM21 DATASET

Networks	Training rates	
	50%	80%
GoogLeNet [4]	92.70±0.60	94.31±0.89
VGG-VD-16 [4]	94.14±0.69	95.21±1.20
CaffeNet [4]	93.98±0.67	95.02±0.81
TEX-Net-LF [42]	96.91±0.36	97.72±0.54
D-CNN [65]	-	98.93±0.10
ARCNet [43]	96.81±0.14	99.12±0.40
ACNet [44]	-	**99.76±0.10**
MG-CAP [66]	-	99.0±0.10
SF-CNN [45]	-	99.05±0.27
FACNN [34]	-	98.81±0.24
SAFF [50]	-	97.02±0.78
SCCov [30]	-	99.05±0.25
EfficientNet-B3-aux [21]	98.22±0.49	99.09±0.17
Fine-tuned ResNet	97.33±0.21	97.74±0.33
Fine-tuned Oct-ResNet	98.10±0.28	98.67±0.26
MF^{2}CNet	**98.76±0.18**	99.52±0.25

we have also carried out experiments on fine-tuned ResNet and fine-tuned Oct-ResNet on the same conditions as MF^{2}CNet.

From the experimental results in Table II, it can be seen that the classification performance of MF^{2}CNet on the UCM21 dataset is better than ones of almost other methods. The OA of ACNet is higher than the one of MF^{2}CNet by about 0.24% (OA = 99.52%). In fact, the classification accuracy of the proposed network is only slightly lower than ACNet for the UCM21 dataset and has better classification performance for other datasets, especially AID30 and NWPU45. Compared with fine-tuned Oct-ResNet and fine-tuned ResNet, MF^{2}CNet increases by 0.85% and 1.78% (Tr = 80%), respectively. However, the proposed network yields the best classification results (OA = 98.76%) when Tr is 50%. The CM is used to reflect the classification performance of the proposed network further. The CMs of fine-tuned ResNet, fine-tuned Oct-ResNet, and MF^{2}CNet are shown in Fig. 9. Overall, MF^{2}CNet achieves better classification performance than fine-tuned ResNet and fine-tuned Oct-ResNet. In all CMs, the *medium residential* is the most misclassified category, wrongly identified as *dense residential* and *sparse residential* categories. The reason for the misclassification is that the scenes of the three classes all contain residential houses, while the classification accuracies on 13 classes of the UCM21 dataset achieve 100% by using MF^{2}CNet.

Experiment 2 (RSSCN7 Dataset): Second, we do experiments on the RSSCN7 dataset, which is widely utilized to verify network performance in RSSC because of its high intraclass variability, small interclass differences, and relatively large scales. Same as [61], the Trs of the RSSCN7 are set to 20% and 50%, respectively. Four approaches are compared with MF^{2}CNet, and the OAs (%) of the RSSCN7 dataset are shown in Table III. It can be seen that the classification performance of MF^{2}CNet is better than other methods. The OAs of MF^{2}CNet achieve 93.16% (Tr = 20%) and

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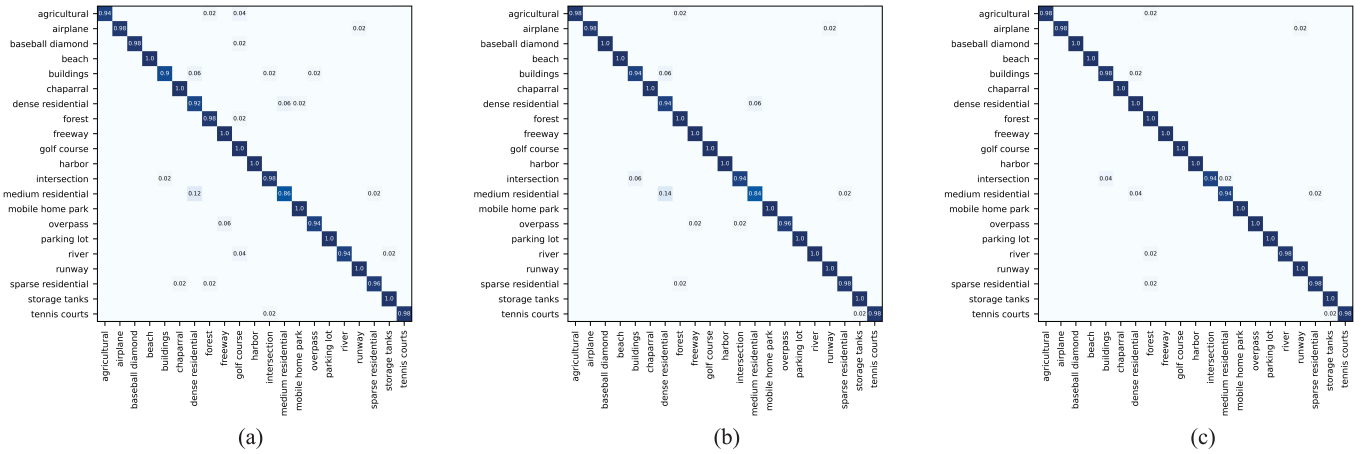
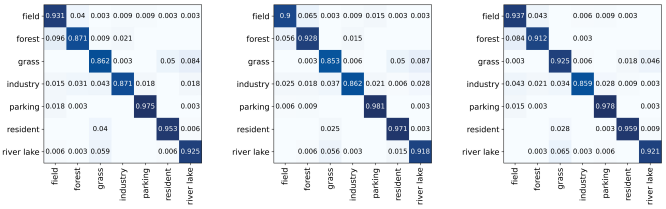


(a) (b) (c)

Fig. 9. Comparison of CMs on the UCM21 dataset with Tr = 50%. (a) Fine-tuned ResNet. (b) Fine-tuned Oct-ResNet. (c) MF^{2}CNet.

TABLE III
COMPARISON OF OAs (%) ON THE RSSCN7 DATASET

Networks	Training rates	
	20%	50%
GoogLeNet [4]	82.55±1.11	85.84±0.92
VGG-VD-16 [4]	83.98±0.87	87.18±0.94
CaffeNet [4]	85.57±0.95	88.25±0.62
TEX-Net-LF [42]	92.45±0.45	94.00±0.57
Fine-tuned ResNet	90.57±0.21	92.15±0.33
Fine-tuned Oct-ResNet	92.05±0.28	94.04±0.36
MF^{2}CNet	**93.16±0.19**	**95.64±0.26**



(a) (b) (c)

Fig. 10. Comparison of CMs on the RSSCN7 dataset with Tr = 20%. (a) Fine-tuned ResNet. (b) Fine-tuned Oct-ResNet. (c) MF^{2}CNet.

95.64% (Tr = 50%). When Tr is 20%, the OAs of MF^{2}CNet has 0.71% higher than TEX-Net-LF and 7.59% higher than CaffeNet. Granted, the performance of the proposed method is better than that of these networks. When Tr is 50%, the proposed network can also achieve good scene classification performance. For the RSSCN7 dataset, the CMs are carried out under the condition of Tr = 20%. The CMs of fine-tuned ResNet, fine-tuned Oct-ResNet, and MF^{2}CNet are shown in Fig. 10. We can see that MF^{2}CNet has the best classification performance. Most classification errors are generated in the *grass* category.

Experiment 3 (AID30 Dataset): Third, we do experiments on the AID30 dataset, which is widely used in RSSC tasks due to the large intraclass variation, small interclass dissimilarity, and the great variation in scale range. Same as [62], the Trs are

TABLE IV
COMPARISON OF OAs (%) ON THE AID30 DATASET

Networks	Training rates	
	20%	50%
GoogLeNet [4]	83.44±0.40	86.39±0.55
VGG-VD-16 [4]	86.59±0.29	89.64±0.36
CaffeNet [4]	86.86±0.47	89.53±0.31
TEX-Net-LF [42]	93.81±0.12	95.73±0.16
D-CNN [65]	90.82±0.16	96.89±0.10
ARCNet [43]	88.75±0.40	93.10±0.55
ACNet [44]	93.33±0.29	95.38±0.29
MG-CAP [66]	93.34±0.18	96.12±0.12
SF-CNN [45]	93.60±0.12	96.66±0.11
FACNN [34]	-	95.45±0.11
SAFF [50]	90.25±0.29	93.83±0.28
SCCov [30]	93.12±0.25	96.10±0.16
EfficientNet-B3-aux [21]	94.19±0.15	96.56±0.14
Fine-tuned ResNet	92.66±0.41	93.73±0.33
Fine-tuned Oct-ResNet	94.35±0.38	95.38±0.36
MF^{2}CNet	**95.54±0.17**	**97.02±0.28**

set to 20% and 50%, respectively. Thirteen currently proposed approaches are chosen to compare with MF^{2}CNet, as shown in Table IV. It can be seen that the classification accuracies of MF^{2}CNet achieve 95.54% (Tr = 20%) and 97.02% (Tr = 50%). When Tr is 20%, MF^{2}CNet is 2.42% higher than SCCov and 4.72% higher than D-CNN. It is obvious that the performance of MF^{2}CNet is better than that of these networks. For the AID30 dataset, the CMs are carried out when Tr is set to 20%. The CMs of fine-tuned ResNet, fine-tuned Oct-ResNet, and MF^{2}CNet are shown in Fig. 11. We can see that MF^{2}CNet has the best classification performance and three categories with a classification accuracy of 100%. The *center* is the category with the most classification errors in the three models, and the classification accuracies are 71.6%, 75.4%, and 80.7%, respectively. The *center* category is mistakenly identified as *square* and *church* categories. The reason is that they all have similar spatial structures and architectural layouts. Nevertheless, our network can still achieve better

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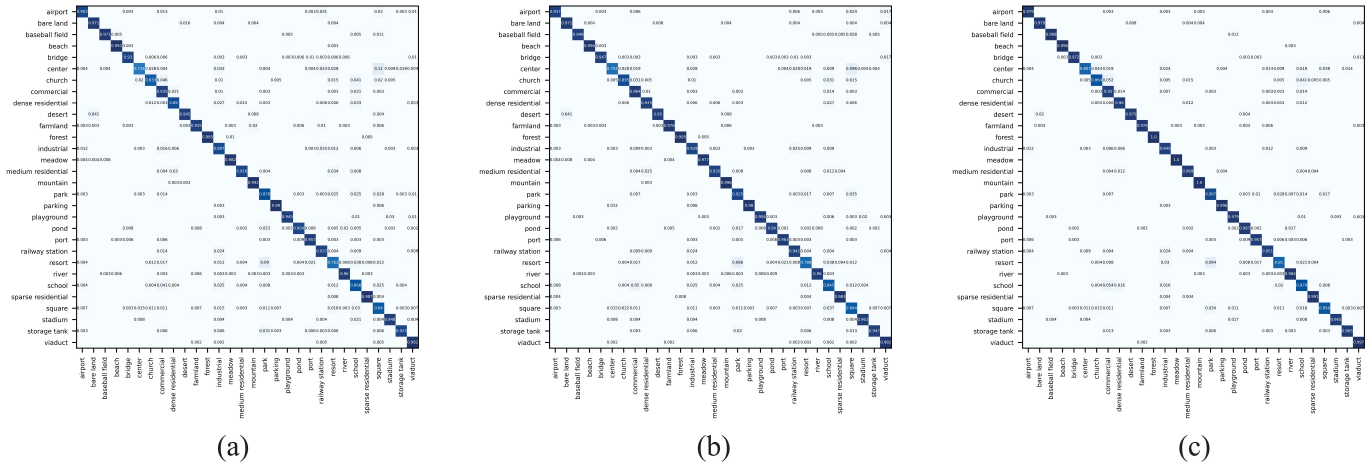

(a) (b) (c)

Fig. 11. Comparison of CMs on the AID30 dataset with Tr = 20%. (a) Fine-tuned ResNet. (b) Fine-tuned Oct-ResNet. (c) MF^{2}CNet.

TABLE V
COMPARISON OF OAS (%) ON THE NWPU45 DATASET

Networks	Training rates	
	10%	20%
GoogLeNet [4]	76.19±0.38	78.48±0.26
VGG-16 [4]	76.48±0.18	79.79±0.15
AlexNet [4]	76.69±0.21	79.85±0.13
D-CNN [65]	89.22±0.50	91.89±0.22
ACNet [44]	91.09±0.13	92.42±0.16
MG-CAP [66]	90.83±0.12	92.95±0.13
SF-CNN [45]	89.89±0.16	92.55±0.14
SAFF [50]	84.38±0.19	87.86±0.14
SCCov [30]	89.30±0.35	92.10±0.25
EfficientNet-B3-aux [21]	91.08±0.14	93.81±0.07
Fine-tuned ResNet	89.23±0.42	90.87±0.38
Fine-tuned Oct-ResNet	90.99±0.36	92.38±0.29
MF^{2}CNet	**92.07±0.22**	**93.85±0.27**

classification performance than other networks on the AID30 dataset.

Experiment 4 (NWPU45 Dataset): Then, we do experiments on the NWPU45 dataset, which is used as a large-scale remote sensing dataset in RSSC tasks because of its rich image changes, high intraclass diversity, and strong interclass similarity. Same as [4], the Trs of the NWPU45 dataset are set to 10% and 20%, respectively. Ten currently proposed approaches are chosen to compare with MF^{2}CNet, as shown in Table V. We can see that the classification performance of MF^{2}CNet achieves 92.07% (Tr = 10%) and 93.85% (Tr = 20%). When Tr is 10%, MF^{2}CNet is 2.77% higher than SCCov and 2.85% higher than D-CNN. It is obvious that the performance of MF^{2}CNet is better than that of these networks. When Tr is 20%, the OAs of MF^{2}CNet are also the highest. We carried out CMs of the NWPU45 dataset when Tr is set to 10%. Comparisons of CMs with fine-tuned ResNet, fine-tuned Oct-ResNet, and MF^{2}CNet are shown in Fig. 12. We can see that MF^{2}CNet has the best classification effect, and the classification effect of fine-tuned ResNet is the worst. In the CMs of fine-tuned ResNet, fine-tuned Oct-ResNet, and MF^{2}CNet, the *church* category is the category with the most classification errors, and the classification accuracies are 61.9%, 71.9%, and 74.1%, respectively. The *church* category is mistakenly identified as *palace* category. The reason is that *church* and *palace* categories contain similar buildings and background information. However, MF^{2}CNet can still achieve higher classification accuracy in *church* and *palace* categories.

V. DISCUSSION

In this part, we select MF^2 module to fuse the features of different stages to study the impact of multiscale feature extraction on RSSC. We discuss the influence of high- and low-frequency features in the MF^2 module. In addition, we also study the effects of Oct Conv, MF^2, and GCP on the classification performance of the proposed MF^{2}CNet and carried out relevant ablation experiments. Finally, we use the visualization method Grad-CAM++ [67] to visualize the classification results. AID30 and NWPU45 datasets are selected, and the experimental setup is the same as in Section IV. To avoid any bias induced by random sampling, the classification results of the experiments are averaged over five trials with standard deviation reported.

A. Selection of Different Stage in MF^2 Module

We design the MF^2 module to fuse the features of different stages to study the impact of multiscale features. Seven different networks are designed for comparative experiments. They are only fusing the features of one stage (stage 2/3/4), fusing the features of two stages (stages 2 and 3, stages 2 and 4, and stages 3 and 4), or fusing the features of three stages (MF^{2}CNet). In order to make the comparative experiments fair, we control the channel number of the output feature maps of Conv 2D 1 × 1 to make the number of feature map's channels output by the MF^2 module of all networks to be equal. The experimental results of different stages of MF^2 module selection are shown in Table VI.

In Table VI, for Network A, B, or C that only employs the features of one stage, the classification accuracy of Network C

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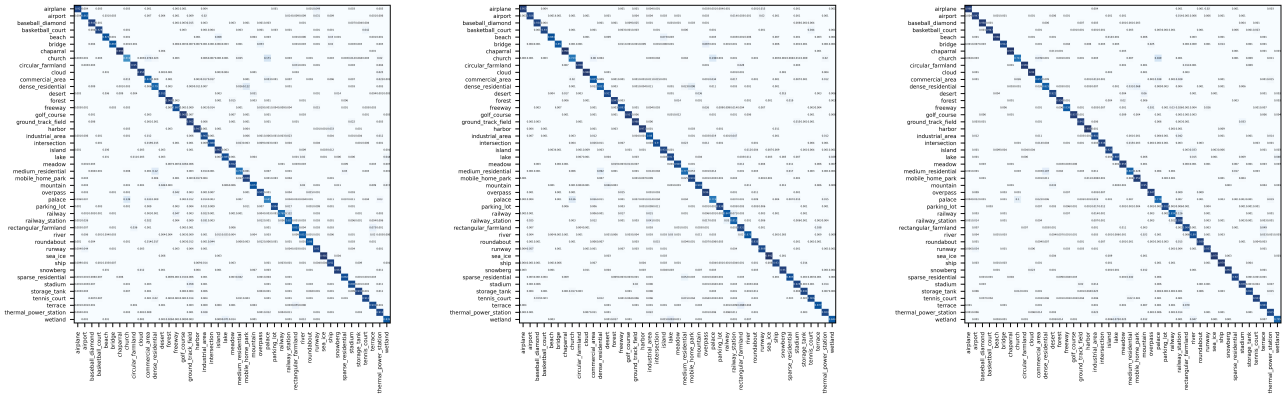

(a) (b) (c)

Fig. 12. Comparison of CMs on the NWPU45 dataset with Tr = 10%. (a) Fine-tuned ResNet. (b) Fine-tuned Oct-ResNet. (c) MF^{2}CNet.

TABLE VI

EXPERIMENTAL RESULTS OF DIFFERENT STAGES OF MF^2 MODULE SELECTION. AID30 AND NWPU45 ARE SELECTED FOR THE EXPERIMENTS. STAGE 2, STAGE 3, AND STAGE 4 ARE REPRESENTED BY S2, S3, AND S4, RESPECTIVELY

Methods	S2	S3	S4	AID30				NWPU45			
				Tr = 20%	Tr = 50%	Parameters	FLOPs	Tr = 10%	Tr = 20%	Parameters	FLOPs
Network A	✓			82.66 ± 0.57	83.91 ± 0.51	7.77M	0.94G	78.36 ± 0.42	80.89 ± 0.65	10.84M	0.95G
Network B		✓		86.35 ± 0.42	87.69 ± 0.50	15.03M	1.64G	81.99 ± 0.36	83.26 ± 0.51	18.10M	1.64G
Network C			✓	94.43 ± 0.46	95.79 ± 0.41	30.98M	2.46G	91.03 ± 0.39	92.86 ± 0.40	34.05M	2.46G
Network D	✓	✓		88.21 ± 0.55	89.36 ± 0.36	14.95M	1.63G	84.97 ± 0.61	87.04 ± 0.45	18.02M	1.64G
Network E	✓		✓	95.08 ± 0.31	96.30 ± 0.29	30.22M	2.42G	91.59 ± 0.36	93.17 ± 0.29	33.30M	2.42G
Network F		✓	✓	95.23 ± 0.23	96.78 ± 0.34	30.33M	2.43G	91.80 ± 0.17	93.58 ± 0.15	33.41M	2.43G
MF^{2}CNet	✓	✓	✓	**95.54 ± 0.17**	**97.02 ± 0.28**	30.12M	2.42G	**92.07 ± 0.22**	**93.85 ± 0.27**	33.20M	2.42G

is higher than that of Network B, and the accuracy of Network B is higher than that of Network A. The trend of parameters and FLOPs of the three networks is opposite to that of classification accuracies because Networks A and B are truncated, and some later stages are not used. There are slight differences among the parameters and FLOPs of Network E, Network F, and MF^{2}CNet, which is caused by MF^2 modules of the three networks that integrate the feature maps of different stages. Different numbers of channels of these feature maps lead to the different parameters and FLOPs of the three networks. By integrating feature maps from two stages, Networks E and F can achieve better classification performance compared with Network C. From this, we can see that the classification performance can be effectively improved by fusing feature maps of multiple stages. In addition, the classification accuracy of MF^{2}CNet with complete feature fusions of three stages is the highest. Considering the complexity and classification accuracy, the proposed MF^{2}CNet can achieve satisfactory classification performance. Therefore, the MF^2 module can improve the classification performance of the proposed network for RSSC by fusing multiscale features.

B. Selection of High- and Low-Frequency Features in MF^2 Module

In MF^{2}CNet, the high- and low-frequency features of stages 2–4 are fused by the MF^2 module. However, it is uncertain which is more important for classification performance. Therefore, we design experiments by selecting high-frequency features, low-frequency features, or both of them as fusion features of the MF^2 module. The results of the comparative experiments are shown in Table VII.

To investigate the influence of high-frequency features on the classification results of MF^{2}CNet, we remove the low-frequency features branches of stage 2 and stage 3 in the MF^2 module and only retain the features of stage 4 and the high-frequency branches of stage 2 and stage 3. At the same time, to ensure the fairness of the comparison, we expand the number of $\mathbf{H1}''$ and $\mathbf{H2}''$ channels by 1×1 convolution and name the network as HF-MF^{2}CNet, in which the number of parameters and FLOPs are set as the same as MF^{2}CNet. Similarly, only using low-frequency features will also reduce the classification accuracies of MF^{2}CNet. We retain the features of stage 4 and the low-frequency features of stage 2 and stage 3. To ensure the fairness of the comparison, we expand the number of $\mathbf{L1}''$ and $\mathbf{L2}''$ channels by 1×1 convolution and name the network as LF-MF^{2}CNet, in which the number of parameters and FLOPs are also set as the same as MF^{2}CNet.

In Table VII, taking the results of the NWPU45 dataset as an example, we can see that the OA of HF-MF^{2}CNet only using high-frequency features and the OA of LF-MF^{2}CNet only using low-frequency features are 0.75% and 0.87% lower than that of MF^{2}CNet, respectively. However, the OAs of HF-MF^{2}CNet and LF-MF^{2}CNet are 0.33% and 0.21% higher than that of fine-tuned Oct-ResNet, respectively. By comparing the experimental results, it can be seen that the classification

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TABLE VII

RESULTS OF THE COMPARATIVE EXPERIMENT. AID30 AND NWPU45 ARE SELECTED FOR THE EXPERIMENTS. HIGH- AND LOW-FREQUENCY ARE REPRESENTED BY HF AND LF, RESPECTIVELY

Methods	HF	LF	AID30				MWPU45			
			Tr = 20%	Tr = 50%	Parameters	FLOPs	Tr = 10%	Tr = 20%	Parameters	FLOPs
Fine-tuned Oct-ResNet			94.35 ± 0.38	95.38 ± 0.36	23.57M	2.39G	90.99 ± 0.36	92.38 ± 0.29	23.60M	2.39G
HF-MF^{2}CNet	✓		94.62 ± 0.46	96.14 ± 0.28	30.12M	2.42G	91.32 ± 0.41	93.09 ± 0.37	33.20M	2.42G
LF-MF^{2}CNet		✓	94.37 ± 0.52	95.76 ± 0.47	30.12M	2.42G	91.20 ± 0.38	92.73 ± 0.31	33.20M	2.42G
MF^{2}CNet	✓	✓	**95.54 ± 0.17**	**97.02 ± 0.28**	30.12M	2.42G	**92.07 ± 0.22**	**93.85 ± 0.27**	33.20M	2.42G

TABLE VIII

RESULTS OF THE ABLATION EXPERIMENTS. AID30 AND NWPU45 ARE SELECTED FOR THE EXPERIMENTS. OCT CONV IS REPRESENTED BY OCT. THE MF^2 MODULE IS REPRESENTED BY MF^2

Methods	Oct	MF^2	GCP	AID30				MWPU45			
				Tr = 20%	Tr = 50%	Parameters	FLOPs	Tr = 10%	Tr = 20%	Parameters	FLOPs
Fine-tuned ResNet				92.66 ± 0.41	93.73 ± 0.33	23.57M	4.12G	89.23 ± 0.42	90.87 ± 0.38	23.60M	4.12G
Fine-tuned Oct-ResNet	✓			94.35 ± 0.38	95.38 ± 0.36	23.57M	2.39G	90.99 ± 0.36	92.38 ± 0.29	23.60M	2.39G
MF^{2}CNet with GAP	✓	✓		94.30 ± 0.52	95.07 ± 0.49	23.99M	2.41G	90.78 ± 0.43	91.89 ± 0.38	24.00M	2.41G
MF^{2}CNet without MF^2	✓		✓	94.43 ± 0.46	95.79 ± 0.41	30.98M	2.46G	91.03 ± 0.39	92.86 ± 0.40	34.05M	2.46G
MF^{2}CNet	✓	✓	✓	**95.54 ± 0.17**	**97.02 ± 0.28**	30.12M	2.42G	**92.07 ± 0.22**	**93.85 ± 0.27**	33.20M	2.42G

performance of the feature fusion methods using only high-frequency features and only low-frequency features is worse than the classification performance of the proposed MF^{2}CNet. This reflects the feasibility of the proposed MF^{2}CNet and shows that high- and low-frequency features are both very important in the proposed MF^{2}CNet for RSSC tasks.

C. Ablation Experiments

We conducted the ablation experiments to test the effect of three modules in MF^{2}CNet. AID30 and the NWPU45 datasets are selected to verify the effectiveness of these methods. As large RSI datasets, these two datasets can reflect the classification performance of the network. We designed five networks: fine-tuned ResNet, fine-tuned Oct-ResNet, MF^{2}CNet with GAP, MF^{2}CNet without MF^2, and MF^{2}CNet. The results of the ablation experiments are shown in Table VIII.

1) Evaluation of Oct Conv: To evaluate the effect of Oct Conv in MF^{2}CNet, we compare the experimental results of fine-tuned ResNet and fine-tuned Oct-ResNet. As shown in Table VIII, the OA of fine-tuned Oct-ResNet on the NWPU45 dataset (Tr = 10%) is 1.76% higher than that of fine-tuned ResNet, which indicates that Oct Conv can optimize ResNet. The reason why Oct-ResNet can achieve better classification performance than ResNet is that Oct Conv fuses low-frequency features with high-frequency features of ResNet, which can express richer information. Moreover, the addition of Oct Conv also greatly reduces the FLOPs of the network. However, Oct-ResNet does not make full use of the high- and low-frequency features from the middle layers, which may lose some useful information. Therefore, the MF^2 module is designed to fuse the multiscale features from different layers.

2) Evaluation of MF^2 Module: To evaluate the impact of the MF^2 module in MF^{2}CNet, MF^{2}CNet without MF^2 is compared with the proposed MF^{2}CNet. Specifically, the channels of $\mathbf{F3}''$ are expanded by five times using 1×1 convolution to make the ablation experiments as fair as possible. Because the MF^2 module leads to different parameters of the network, the utilization of the MF^2 module can slightly reduce the parameters and FLOPs of the network. In Table VIII, it can be seen that the OA of MF^{2}CNet is 1.04% higher than that of MF^{2}CNet without MF^2. Therefore, the MF^2 module is significant for MF^{2}CNet since it can extract multiscale features from RSIs.

3) Evaluation of GCP Module: To evaluate the effect of the GCP module in MF^{2}CNet, we design a network called MF^{2}CNet with GAP, which uses GAP instead of GCP and is compared with the proposed MF^{2}CNet. In Table VIII, the OA of MF^{2}CNet is 1.29% higher than that of MF^{2}CNet with GAP. The reason is that only the first-order information (mean) is retained by GAP, and then, multifrequency and multiscale features extracted by the MF^2 module are not fully utilized, which leads to the degradation of classification performance. The difference between MF^{2}CNet without MF^2 and MF^{2}CNet in parameters is due to the GCP module and the full connection layer, but, after we balance the accuracy and parameters, we can see that the GCP module is beneficial to the network. This is because GCP utilizes the second-order (covariance) information, which can get more distinguishable features for RSSC tasks.

Through the analysis of three factors, it can be seen that Oct Conv can improve the classification performance of the network. The separate use of the GCP or MF^2 module is unsatisfactory, but their collaboration works effectively for RSSC. Therefore, we fully consider the characteristics of Oct Conv, MF^2, and GCP modules and combine them to improve the classification performance of the designed network.

D. Visualization Experiments

For the purpose of better explaining the classification mechanism of the proposed network on RSSC, the Grad

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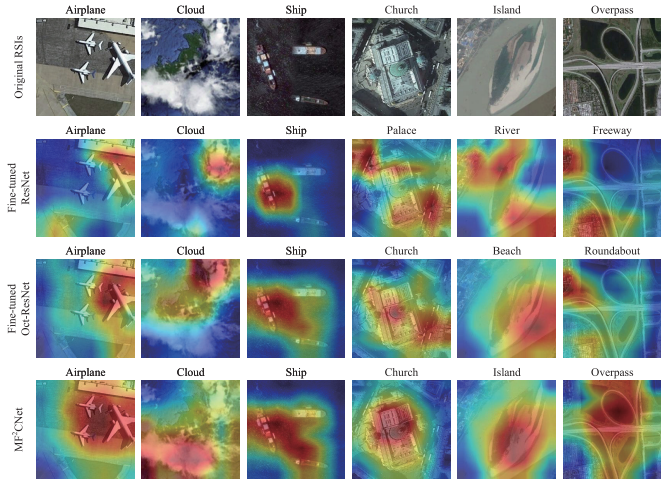

Fig. 13. Visualization results generated by Grad CAM++. Original RSIs (first line), visualization results generated by fine-tuned ResNet (second line), visualization results generated by fine-tuned Oct-ResNet (third line), and visualization results generated by MF^{2}CNet (last line). The descriptions on the top of the original RSIs are the true labels, while the descriptions on the top of visualization results are the prediction labels.

CAM++ [67] method is used to visualize the attention position of objects in RSIs. We use the feature maps from the last convolution layer for visualization. In Fig. 13, six RSIs with variable objects are chosen from the NWPU45 dataset to reflect the representation ability of our network for multiscale features.

As we can see from the visualization and prediction results in Fig. 13, fine-tuned ResNet identified three scene categories, fine-tuned Oct-ResNet identified four scene categories, and the proposed MF^{2}CNet identified all the selected scene categories. For MF^{2}CNet, we find that the attention positions are more accurate, and the attention ranges are wider than ones of fine-tuned ResNet and fine-tuned Oct-ResNet. For the *airplane* category, the attention ranges using fine-tuned ResNet just cover large airplanes, but the small airplanes next to them are missed. However, MF^{2}CNet can detect the large and small airplanes, and then, the semantic information of the *airplane* category can be extracted completely. For the *overpass* category, the attention range of ResNet and Oct-ResNet cannot well reflect the location of *overpass*, while MF^{2}CNet can still perform well for the complex scenes by capturing multiscale and multifrequency features and using GCP to retain second-order information.

VI. Conclusion

In this article, we proposed an MF^{2}CNet, which can extract multifrequency and multiscale features for RSSC. In MF^{2}CNet, the MFE module is used to extract multifrequency features with Oct Conv. Then, the features of different layers in the network are fused by the MF^2 module. To capture richer statistics of multifrequency and multiscale features, GCP is used to extract the second-order information of RSIs. The results of experiments show that high- and low-frequency features are favorable for classification accuracy. The results of ablation experiments show that Oct Conv, the MF^2 module, and GCP are beneficial to the improvement of the network. At the same time, by comparing with many state-of-the-art networks, we can see the improved performance of the proposed MF^{2}CNet in RSSC tasks.

In future work, we will explore representation learning methods to extract more valuable features of RSIs and study deep networks from the perspective of physics-informed neural networks for RSSC.

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